

CoSimComponent *(abstract base class)*

Typed Port Interface

Input
Ports

PortSpec
type, direction, unit

defines
per port

PortState
value, time stamp

Output
Ports

Lifecycle

1. Construction
2. Configuration
3. Initialization
4. Simulation Loop
 - 4.1 Set Inputs
 - 4.2 Do Step
 - 4.3 Read Outputs
5. Reset / Release

State Management

Physical State

Inspection and
State Transfer

Rollback

Event
Localization

Model Structure

Input-Output Dependencies
Direct Feedthrough

Hybrid Capabilities

Event Indicators
Event Subscriptions

Legend: dashed border = optional capability